Advanced Data Analytics Summary Report

National Retail Company: Business Intelligence & Interactive Dashboard Development

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Data Preparation & Exploration

I used transform data to access power query where I used to enable show whitespaces, column distribution, column quality and column quantity to view my data. But I didn't have null or missing values since I already covered that in my week 2 by handling all missing values and replaced them using 0, these values were in total sales. I also went on to analyze my data further by listing all the fields needed for my analysis and ensured all the data types were correct and data format. I went on to create date table that will format my date to correct format that I can use when preparing my previous year's sales and profit.

Sales Performance Analysis

Total sales for a period of 3years is 2.30M. With the last month of the year having spikes over years stated and January & February being the least performing months across the 3years period (2015-2017).

Profit Performance Analysis

Total profit generated from 2016-2017 is $286.40k and profit margin of 12.47% with profit following the same trend of sales by having spikes on December and the 1st quarter underperforming. However, the last quarter accounts for a disproportional high percentage of annual profit, creating cashflow concentration risk.

Product Performance

Technology dominates both profit and sales metrics driven heavily by phones. With copiers generating high profit margin despite low sale volume than phones. Furniture on the other hand is the driver of loss with tables generating loss of approximately $33600. This brought us to the conclusion that high sales don't usually equal high profits.

Region Performance

West region has proven to be the clear leader based on profit of approximately $108.42k driven by the state of California while central region generates profit of approximately $39.70k with Texas being the main driver of loss with loss of over $25.73k which is primary attributed to deep discounting on Binder's subcategory and low performance on office supplies.

Customer & Segment

The most performing customer segment is consumers, which account for close to 52% of sales and home office being the least performing customer segment. The heavy reliance on consumer segments exposes the business to high risk when consumer spending declines.

Discount Analysis

There's negative correlation between high discounts and profitability because areas with high discounts have relatively low profits. For example, Furniture has average discount of approximately 26% which is the primary driver of loss and Phones on the other hand has average discount of 15% and it's the main driver of profit. Analysis indicates that excessive discounts are leaking potential revenue, particularly in the South & Central region.